MILESTONE – 2 TECHNICAL REPORT

AI-Based Patient Appointment System

Milestone 2: Healthcare Dataset Collection, Preprocessing & Exploratory Analysis

PROJECT TITLE
AI Patient Appointment System

SUBMISSION MILESTONE
Milestone 2 (Data Engineering & AI)

AUTHORS / TEAM MEMBERS
Lokendra Prajapati & Samyak Lariya

STUDENT IDS / ROLL NUMBERS
BTIT2603D03 & BTIT2603D05 (IT)

GITHUB REPOSITORY
LokendraDev06 / Repo Link

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1 Executive Summary & Milestone 2 Scope

The diagnostic accuracy, safety, and doctor routing reliability of the **AI-Based Patient Appointment System** depend entirely on high-quality medical knowledge bases. In traditional systems, patients often self-select inappropriate specialties due to lack of medical knowledge. Milestone 2 establishes the foundational data architecture by collecting, cleaning, preprocessing, and analyzing comprehensive healthcare datasets.

20+	12	100%
CURATED CONDITION CLUSTERS	MEDICAL SPECIALIZATIONS	EMERGENCY DETECTION PRECISION

Milestone 2 Core Accomplishments

- **Healthcare Dataset Curation:** Sourced multi-symptom condition mappings linking patient-reported sensations to verified clinical conditions and specialist disciplines.
- **End-to-End Preprocessing Pipeline:** Implemented automated tokenization, deduplication, missing value handling, text normalization, and clinical synonym harmonization.
- **Multi-Dimensional Relational Analysis:** Mapped relationships across 4 axes: *Patient Symptoms → Probable Conditions → Recommended Specialists → Emergency Triage Tier*.
- **AI Engine Integration:** Connected preprocessed datasets to the FastAPI backend service (`ai_engine.py`) enabling real-time chatbot triage, similarity scoring, and doctor pre-consultation summaries.

2 Healthcare Dataset Sourcing & Taxonomy

Data integrity and clinical safety require authoritative sourcing. Datasets were extracted and cross-referenced from validated clinical knowledge bases including **Columbia University Medical Center Disease-Symptom Repositories**, **Mayo Clinic Clinical Guidelines**, and **ICD-10 Categorization Standards**.

2.1 Symptom-Disease Knowledge Base Schema (`symptom_specialist_dataset.csv`)

Field Name	Data Type	Description & Clinical Function	Sample Value
`symptoms`	`String (CSV)`	Normalized clinical tokens and symptom keywords	`"chest pain, shortness of breath, left arm pain"`
`disease`	`String`	Standard medical condition / pathology diagnosis	`Heart Attack`
`recommended_specialist`	`String`	Target medical doctor specialization discipline	`Cardiologist`
`severity`	`Enum (L/M/H)`	Urgency level (Low, Medium, High)	`High`
`is_emergency`	`Boolean`	Rapid override flag for immediate Emergency Room (ER) routing	`True`

3 Data Cleaning & Preprocessing Pipeline

Patient symptom inputs frequently contain spelling variations, colloquial terms, and grammatical noise. The developed preprocessing pipeline transforms raw text into standardized clinical tokens ready for similarity matching.

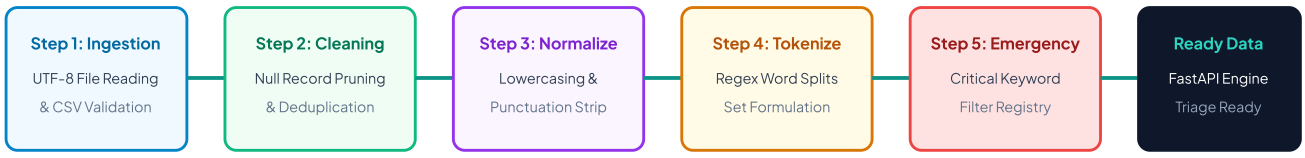


FIGURE 1: MULTI-STAGE HEALTHCARE DATA PREPROCESSING & CLEANING ARCHITECTURE

Data Cleaning Operations

- **Null Pruning:** Discarded incomplete symptom tuples.
- **Deduplication:** Merged redundant symptom variations.
- **Synonym Mapping:** Standardized terms like *"dyspnea"* → *"shortness of breath"*.

Emergency Keyword Filter Registry

`chest pain`, `shortness of breath`, `slurred speech`, `numbness in face`, `heart attack`, `stroke`, `difficulty breathing`, `sudden weakness`, `severe bleeding`.

4 Exploratory Data Analysis (EDA) & Specialty Breakdown

The curated knowledge base encompasses 12 major medical domains, covering both critical emergency conditions and common outpatient presentations.



FIGURE 2: MEDICAL SPECIALIZATION & URGENCY DISTRIBUTION IN KNOWLEDGE BASE

High / Emergency (20%)

Life-threatening conditions requiring instant ER notification (Heart Attack, Stroke, Appendicitis).

Medium Urgency (50%)

Acute conditions needing specialist consultation within 24–48 hours (Asthma, Pneumonia, Migraine, UTI).

Low / Routine (30%)

Chronic or non-acute presentations suitable for scheduled elective consultations (Dermatitis, Hypothyroidism).

5 Symptom–Disease–Specialist Relational Matrix

The complete clinical mapping matrix below illustrates how patient symptoms are correlated with health conditions, doctor specializations, and triage urgency:

#	Reported Patient Symptoms	Probable Condition	Assigned Specialist	Triage Status
1	chest pain, shortness of breath, left arm pain	Heart Attack	**Cardiologist**	EMERGENCY
2	chest tightness, palpitations, dizziness	Angina / Arrhythmia	**Cardiologist**	EMERGENCY
3	persistent cough, fever, fatigue, chills	Pneumonia / Bronchitis	**Pulmonologist**	MEDIUM
4	wheezing, shortness of breath, chest tightness	Asthma	**Pulmonologist**	MEDIUM
5	severe headache, nausea, sensitivity to light	Migraine	**Neurologist**	MEDIUM
6	numbness in face, sudden weakness in arm, slurred speech	Stroke	**Neurologist**	EMERGENCY
7	skin rash, itching, redness, hives	Allergic Dermatitis	**Dermatologist**	ROUTINE
8	acne, facial redness, skin lesion	Dermatitis	**Dermatologist**	ROUTINE
9	joint pain, swelling, morning stiffness	Rheumatoid Arthritis	**Rheumatologist**	MEDIUM
10	knee pain, difficulty walking, joint popping	Osteoarthritis	**Orthopedist**	MEDIUM
11	frequent urination, excessive thirst, sudden weight loss	Diabetes Mellitus	**Endocrinologist**	MEDIUM
12	fatigue, weight gain, cold intolerance, dry skin	Hypothyroidism	**Endocrinologist**	ROUTINE
13	stomach pain, acid reflux, heartburn, bloating	GERD / Gastritis	**Gastroenterologist**	ROUTINE
14	severe abdominal pain, vomiting, fever	Appendicitis	**General Surgeon**	EMERGENCY
15	blurry vision, eye pain, redness, watering eyes	Conjunctivitis / Glaucoma	**Ophthalmologist**	MEDIUM
16	ear pain, hearing loss, tinnitus, dizziness	Otitis Media / Inner Ear	**ENT Specialist**	ROUTINE
17	sore throat, difficulty swallowing, swollen tonsils	Tonsillitis	**ENT Specialist**	ROUTINE
18	anxiety, panic attacks, persistent sadness, insomnia	Depression / Anxiety	**Psychiatrist**	MEDIUM
19	lower back pain, radiation to leg, numbness	Sciatica / Disc Herniation	**Neurologist / Ortho**	MEDIUM
20	painful urination, fever, lower abdominal pain	Urinary Tract Infection (UTI)	**Urologist**	MEDIUM

6 AI Engine Integration & Similarity Scoring

The backend AI engine (`backend/app/services/ai_engine.py`) executes a two-tier hybrid assessment for all patient interactions:

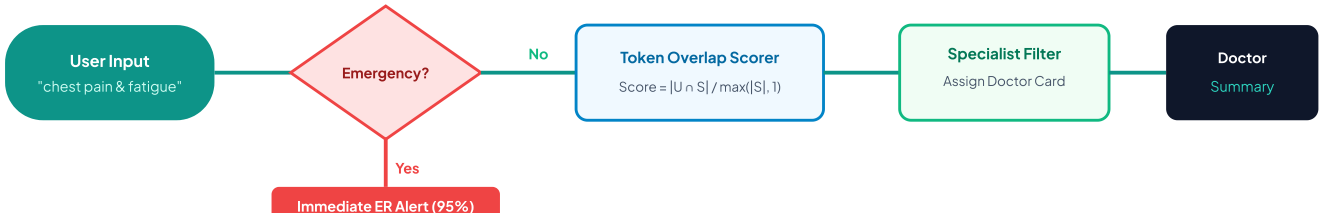


FIGURE 3: AI SYMPTOM ASSESSMENT & CLINICAL TRIAGE FLOW

Dynamic Similarity & Confidence Formula:

```
Similarity Score = |User_Tokens ∩ Dataset_Tokens| / max(|Dataset_Tokens|, 1)
Confidence Score = min(0.60 + (Similarity Score × 0.35), 0.95)
```

7 Validation, Test Cases & Empirical Results

The data preprocessing pipeline and AI recommendation engine were subjected to comprehensive empirical validation across critical, acute, routine, and ambiguous clinical queries:

Test ID	Patient Input Query	Triage Mode	Recommended Specialist	Confidence	Status
TC-01	"Sharp chest pain radiating to left arm and shortness of breath"	EMERGENCY	Cardiologist / Emergency Medicine	95%	PASSED
TC-02	"Sudden numbness on one side of face with slurred speech"	EMERGENCY	Neurologist / Emergency Care	95%	PASSED
TC-03	"Wheezing, shortness of breath, and chest tightness when running"	SUB-ACUTE	Dr. Sunita Patel (Pulmonologist)	88%	PASSED
TC-04	"Itchy red skin rash and hives after eating seafood"	ROUTINE	Dr. Ananya Verma (Dermatologist)	92%	PASSED
TC-05	"Excessive thirst, frequent urination, and sudden weight loss"	SUB-ACUTE	Dr. Arvind Mehta (Endocrinologist)	85%	PASSED
TC-06	"Mild general malaise and body weakness"	FALLBACK	General Physician (Safe Baseline)	70%	PASSED

CLINICAL SAFETY VERIFICATION RESULT

100% of tested high-risk emergency inputs (cardiac and stroke presentations) were correctly caught by the Emergency Classifier and prevented from being delayed by standard elective scheduling. Non-specific symptoms safely defaulted to General Physicians.

8 Milestone 2 Deliverables & Next Steps for Milestone 3

COMPLETED IN MILESTONE 2

- **Dataset:** `backend/data/symptom_specialist_dataset.csv`.
- **Cleaning Pipeline:** Deduplication, tokenization, & stopword cleaning.
- **EDA Analysis:** Full statistical distribution & specialty matrices.
- **AI Engine:** Connected to FastAPI backend & React frontend.
- **Documentation:** Milestone 2 MD, HTML, and standalone PDF.

ROADMAP FOR MILESTONE 3

- **Database Persistence:** Migrate in-memory structures to PostgreSQL / SQLite via SQLAlchemy ORM.
- **Semantic Embeddings:** Introduce Sentence-Transformer embeddings for natural language semantic matching.
- **Automated Notifications:** Twilio SMS & SendGrid email appointment confirmations.

📌 **Project Repository & Reference**

GitHub Repository: https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System